

Laptop Price Prediction

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Table of contents

Introduction	2
Data Source	2
Summary Statistics	2
Missing Values Proportion	7
Missing Pattern	7
Exploratory Data Analysis (EDA)	10
Fix wrong values	10

Introduction

Data Source

The laptop dataset is from Kaggle [Laptop Sales Price Prediction](#).

```
setwd("C:/github_LTY/homework/Final Report") # Set working directory.
laptop <- read.csv("data/laptop.csv", na.strings = c("", "NA"))
```

Because “X” and “Name” are useless, so we delete these columns in advance.

```
laptop <- subset(x = laptop, select = -c(X, Name))
```

Summary Statistics

```
library(Hmisc) # summary statistics
```

Attaching package: 'Hmisc'

The following objects are masked from 'package:base':

format.pval, units

```
latex(describe(laptop), file = "", caption.placement = "top")
```

		laptop	
		27 Variables	1020 Observations
Brand		n missing distinct	
		1020 0 31	
lowest : Acer		Apple	Asus
highest: Ultimus		Walker	Wings
			ASUS
			Xiaomi
			Avita
			Zebronics

Price

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	561	1	82063	59619	25598	31990	43990	63690	94990	149990	199990

lowest : 8000 12990 13990 14490 14990, highest: 450990 453990 489883 519990 599990

Rating

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	17	0.996	4.374	0.2689	4.00	4.05	4.20	4.35	4.55	4.70	4.75

Value	3.95	4.00	4.05	4.10	4.15	4.20	4.25	4.30	4.35	4.40	4.45	4.50	4.55	4.60
Frequency	1	71	56	69	57	66	68	57	71	76	55	57	64	57
Proportion	0.001	0.070	0.055	0.068	0.056	0.065	0.067	0.056	0.070	0.075	0.054	0.056	0.063	0.056

Value	4.65	4.70	4.75
Frequency	53	64	78
Proportion	0.052	0.063	0.076

For the frequency table, variable is rounded to the nearest 0

Processor_brand

n	missing	distinct
1020	0	7

Value	AMD	Apple	HiSilicon	Intel	MediaTek	Microsoft	Qualcomm
Frequency	250	18	1	742	7	1	1
Proportion	0.245	0.018	0.001	0.727	0.007	0.001	0.001

Processor_name

n	missing	distinct
1020	0	36

lowest : AMD Athlon AMD Athlon Pro AMD Athlon Silver AMD Ryzen 3 AMD Ryzen 5
highest: Mediatek MediaTek MediaTek Kompanio Microsoft SQ1 SQ1 Qualcomm X Elite

Processor_variant

n	missing	distinct
996	24	125

lowest : 1005G1 100U 10210U 10210Y 1035G1
highest: N5100 N6000 processor SQ1 Z3735F

Processor_gen

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
891	129	13	0.95	10.45	3.086	5	5	7	12	13	13	13

Value	1	3	4	5	6	7	8	9	10	11	12	13	14
Frequency	1	11	4	92	17	120	7	1	13	113	246	255	11
Proportion	0.001	0.012	0.004	0.103	0.019	0.135	0.008	0.001	0.015	0.127	0.276	0.286	0.012

For the frequency table, variable is rounded to the nearest 0

Core_per_processor

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1008	12	12	0.977	8.572	4.694	2	4	6	8	10	14	16
Value	2	4	5	6	8	10	11	12	14	16	20	24
Frequency	81	118	8	182	198	189	1	93	69	42	3	24
Proportion	0.080	0.117	0.008	0.181	0.196	0.188	0.001	0.092	0.068	0.042	0.003	0.024

For the frequency table, variable is rounded to the nearest 0

Total_processor

n	missing	distinct	Info	Mean	Gmd				
573	447	8	0.902	3.927	2.103				
Value	1	2	4	5	6	8	10	12	
Frequency	7	221	180	1	121	41	1	1	
Proportion	0.012	0.386	0.314	0.002	0.211	0.072	0.002	0.002	

For the frequency table, variable is rounded to the nearest 0

Execution_units

n	missing	distinct	Info	Mean	Gmd
573	447	5	0.734	6.998	2.478
Value	4	6	8	12	16
Frequency	193	3	350	3	24
Proportion	0.337	0.005	0.611	0.005	0.042

For the frequency table, variable is rounded to the nearest 0

Low_Power_Cores

n	missing	distinct	Info	Mean	Gmd
1020	0	2	0.124	0.08627	0.1653
Value	0	2			
Frequency	976	44			
Proportion	0.957	0.043			

Energy_Efficient_Units

n	missing	distinct	Info	Sum	Mean	Gmd
1020	0	2	0.124	44	0.04314	0.08263

Threads

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95		
972	48	14	0.937	12.82	5.988	4	8	8	12	16	20	22		
Value	2	4	5	6	8	12	14	16	18	20	22	24	28	32
Frequency	36	50	1	7	178	346	6	218	11	51	27	15	3	23
Proportion	0.037	0.051	0.001	0.007	0.183	0.356	0.006	0.224	0.011	0.052	0.028	0.015	0.003	0.024

For the frequency table, variable is rounded to the nearest 0

RAM_GB

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	10	0.798	13.99	6.415	8	8	8	16	16	16	32

Value	2	4	8	12	16	18	32	36	48	64
Frequency	1	25	377	2	543	3	62	1	1	5
Proportion	0.001	0.025	0.370	0.002	0.532	0.003	0.061	0.001	0.001	0.005

For the frequency table, variable is rounded to the nearest 0

RAM_type

n	missing	distinct
998	22	12

Value	DDR3	DDR4	DDR5	DDR6	LPDDR3	LPDDR4	LPDDR4X	LPDDR5	LPDDR5X	LPDDR5X4
Frequency	2	522	188	1	1	16	53	173	38	2
Proportion	0.002	0.523	0.188	0.001	0.001	0.016	0.053	0.173	0.038	0.002

Value	PDDR5X	Unified
Frequency	1	1
Proportion	0.001	0.001

Storage_capacity_GB

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	10	0.664	627.7	265	256	512	512	512	512	1000	1000

Value	32	64	128	256	512	1000	1256	1512	2000	4000
Frequency	1	9	19	46	702	218	4	2	18	1
Proportion	0.001	0.009	0.019	0.045	0.688	0.214	0.004	0.002	0.018	0.001

For the frequency table, variable is rounded to the nearest 0

Storage_type

n	missing	distinct
1020	0	4

Value	Hard Disk	NVMe SSD	SSD	Hard Disk & SSD
Frequency	14	1	999	6
Proportion	0.014	0.001	0.979	0.006

Graphics_name

n	missing	distinct
1018	2	136

lowest : 10-Core GPU	10 Core GPU	14 Core GPU	18 Core GPU	30 Core GPU
highest: Nvidia RTX 3050	NVIDIA RTX 4050	NVIDIA RTX A500	NVIDIA RTX NVIDIA T550	Radeon Pro 5500M

Graphics_brand

n	missing	distinct
1018	2	7

Value	Adreno	AMD	Apple	ARM	Intel	NVIDIA	Radeon
Frequency	1	151	18	7	486	354	1
Proportion	0.001	0.148	0.018	0.007	0.477	0.348	0.001

Graphics_GB

n	missing	distinct	Info	Mean	Gmd
368	652	6	0.871	5.94	2.556

Value	2	4	6	8	12	16
Frequency	4	174	87	80	12	11
Proportion	0.011	0.473	0.236	0.217	0.033	0.030

For the frequency table, variable is rounded to the nearest 0

Graphics_integreted

n	missing	distinct
1018	2	2

Value	False	True
Frequency	772	246
Proportion	0.758	0.242

Display_size_inches

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	22	0.869	15.16	1.008	13.5	14.0	14.0	15.6	15.6	16.0	16.1

lowest : 11.6 12 12.4 13 13.3, highest: 16.1 16.2 17 17.3 18

Horizontal_pixel

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	22	0.569	2036	340.2	1366	1920	1920	1920	1920	2560	2880

lowest : 1080 1200 1280 1366 1440, highest: 3024 3072 3200 3456 3840

Vertical_pixel

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	22	0.767	1214	272.7	768	1080	1080	1080	1200	1800	1864

lowest : 768 1024 1080 1200 1280, highest: 2000 2160 2234 2400 2560

ppi

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	62	0.902	157.2	30.93	127.3	141.2	141.2	141.2	161.7	212.3	242.6

lowest : 100.45 111.14 111.94 127.34 129.58, highest: 266.26 266.37 267.08 283.02 337.93

Touch_screen

n	missing	distinct
1020	0	2

Value	False	True
Frequency	904	116
Proportion	0.886	0.114

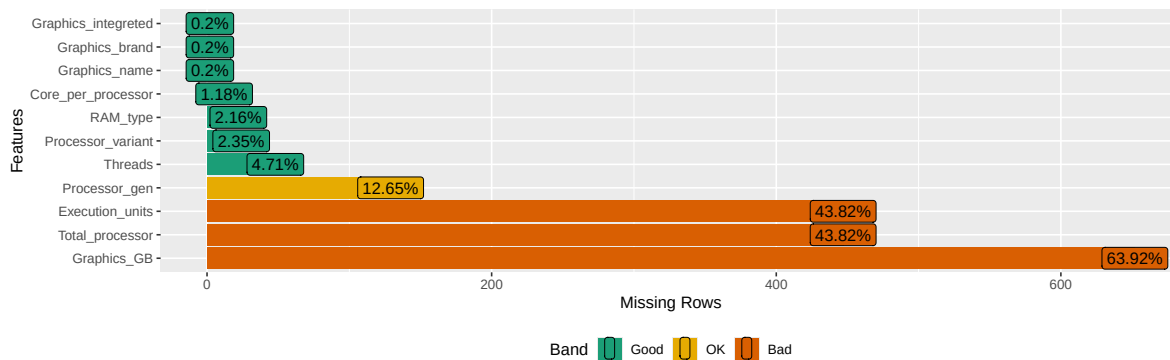
Operating_system

n missing distinct
1020 0 13

lowest : Android 11 OS Chrome OS DOS 3.0 OS DOS OS jio
highest: Mac OS Prime OS Ubuntu OS Windows 10 OS Windows 11 OS

Missing Values Proportion

```
library(DataExplorer)
plot_missing(laptop, missing_only = TRUE)
```



From the figure above, we can discover that Graphics_GB has 63.92% missing values.

Missing Pattern

Next, we want to realize the missing pattern of laptop data.

```
# Customize md.pattern function originally in mice package.
library(mice)
md.pattern.mine <- function(x, plot = TRUE, rotate.names = FALSE, cex = 1.0){
  if(!(is.matrix(x) || is.data.frame(x))){
    stop("Data should be a matrix or dataframe")
  }
  if(ncol(x) < 2){
    stop("Data should have at least two columns")
  }
  R <- is.na(x)
```

```

nmis <- colSums(R)
R <- matrix(R[, order(nmis)], dim(x))
pat <- apply(R, 1, function(x) paste(as.numeric(x), collapse = ""))
sortR <- matrix(R[order(pat), ], dim(x))

if(nrow(x) == 1){mpat <- is.na(x)}
else{mpat <- sortR[!duplicated(sortR), ]}

if (all(!is.na(x))) {
  cat(" /\ \      /\ \n{ `---' } \n{ 0  0 } \n==> V <==")
  cat(" No need for mice. This data set is completely observed.\n")
  cat(" \ \  \ \ /  /\n `-----' \n\n")
  mpat <- t(as.matrix(mpat, byrow = TRUE))
  rownames(mpat) <- table(pat)
}
else{
  if(is.null(dim(mpat))){
    mpat <- t(as.matrix(mpat))
  }
  rownames(mpat) <- table(pat)
}
r <- cbind(abs(mpat - 1), rowSums(mpat))
r <- rbind(r, c(nmis[order(nmis)], sum(nmis)))
if (plot){
  op <- par(mar = rep(0, 4))
  on.exit(par(op))
  plot.new()
  if(is.null(dim(sortR[!duplicated(sortR), ]))) {
    R <- t(as.matrix(r[1:nrow(r) - 1, 1:ncol(r) - 1]))
  }
  else{
    if (is.null(dim(R))) {
      R <- t(as.matrix(R))
    }
  }
  R <- r[1:nrow(r) - 1, 1:ncol(r) - 1]
}
if(rotate.names){
  adj <- c(0, 0.5)
  srt <- 90
  length_of_longest_colname <- max(nchar(colnames(r)))/2.6
  plot.window(xlim = c(-1, ncol(R) + 1),
              ylim = c(-1, nrow(R) + length_of_longest_colname), asp = 1)

```

```

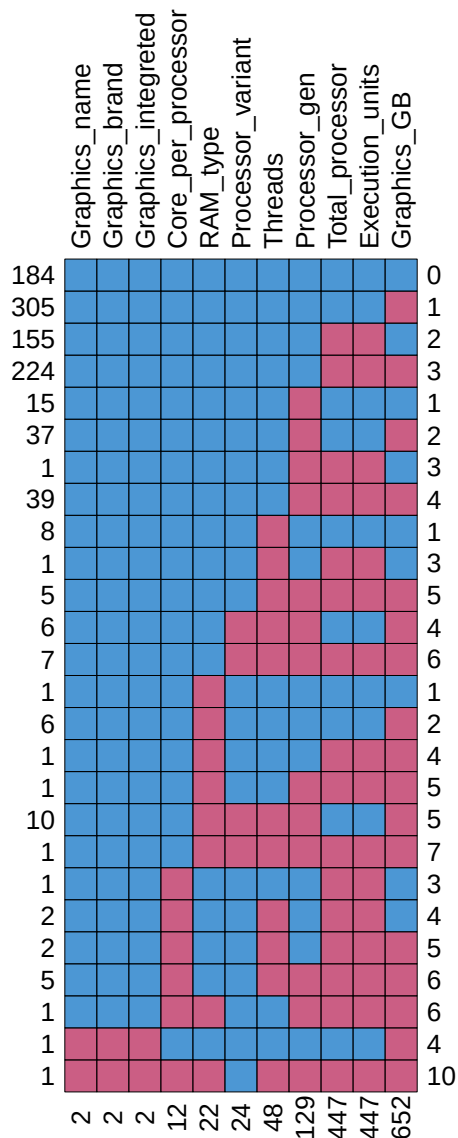
}
else{
  adj <- c(0.5, 0)
  srt <- 0
  plot.window(xlim = c(-1, ncol(R) + 1),
              ylim = c(-1, nrow(R) + 1), asp = 1)
}
M <- cbind(c(row(R)), c(col(R))) - 1
shade <- ifelse(R[nrow(R):1, ], mdc(1), mdc(2))
rect(M[, 2], M[, 1], M[, 2] + 1, M[, 1] + 1, col = shade)
for(i in 1:ncol(R)){
  # Show colnames at the top center.
  text(i - 0.5, nrow(R) + 0.3, colnames(r)[i], adj = adj,
       srt = srt, cex = cex)
  # Show number of missing values for each variable at the bottom center.
  text(i - 0.5, -0.8, nmis[order(nmis)][i], cex = cex, srt = srt)
}
for(i in 1:nrow(R)){
  # Show number of missing values for each subjects at the RHS.
  text(ncol(R) + 0.3, i - 0.5, r[(nrow(r) - 1):1, ncol(r)][i],
       adj = 0, cex = cex)
  # Show number of subjects for each kind of missing pattern at the LHS.
  text(-0.3, i - 0.5, rownames(r)[(nrow(r) - 1):1][i], adj = 1, cex = cex)
}
# show total number of missing entries.
# text(ncol(R) + 1.3, -0.6, r[nrow(r), ncol(r)], cex = cex)
# return(r)
}
else {
  return(r)
}
}

```

```

library(dplyr)
mis_col <- colnames(laptop)[profile_missing(laptop)$num_missing > 0]
laptop[,mis_col] %>%
  md.pattern.mine(rotate.names = T, cex = 2.0)

```



From the missing pattern plot, we can observe that

1. The missing values of “Total_processor” and “Execution_units” occur with each other.

Exploratory Data Analysis (EDA)

Fix wrong values

First, we look at each categorical variables to check whether there are some typing errors.

```
attach(laptop)
sort(table(Brand))
```

Brand

ASUS	Colorful	iBall	Jio	Ninkear	Razer	Tecno	Walker
1	1	1	1	1	1	1	1
AXL	Huawei	Fujitsu	Primebook	Wings	Ultimus	Avita	Gigabyte
2	2	3	3	3	5	6	6
Honor	Xiaomi	LG	Microsoft	Zebronics	Chuwi	Apple	Infinix
6	6	7	7	7	8	20	20
Samsung	Acer	MSI	Dell	Asus	HP	Lenovo	
32	69	97	116	157	213	217	

```
# replace "Asus" with "ASUS".
Brand <- gsub(pattern = "Asus", replacement = "ASUS", x = Brand)
```

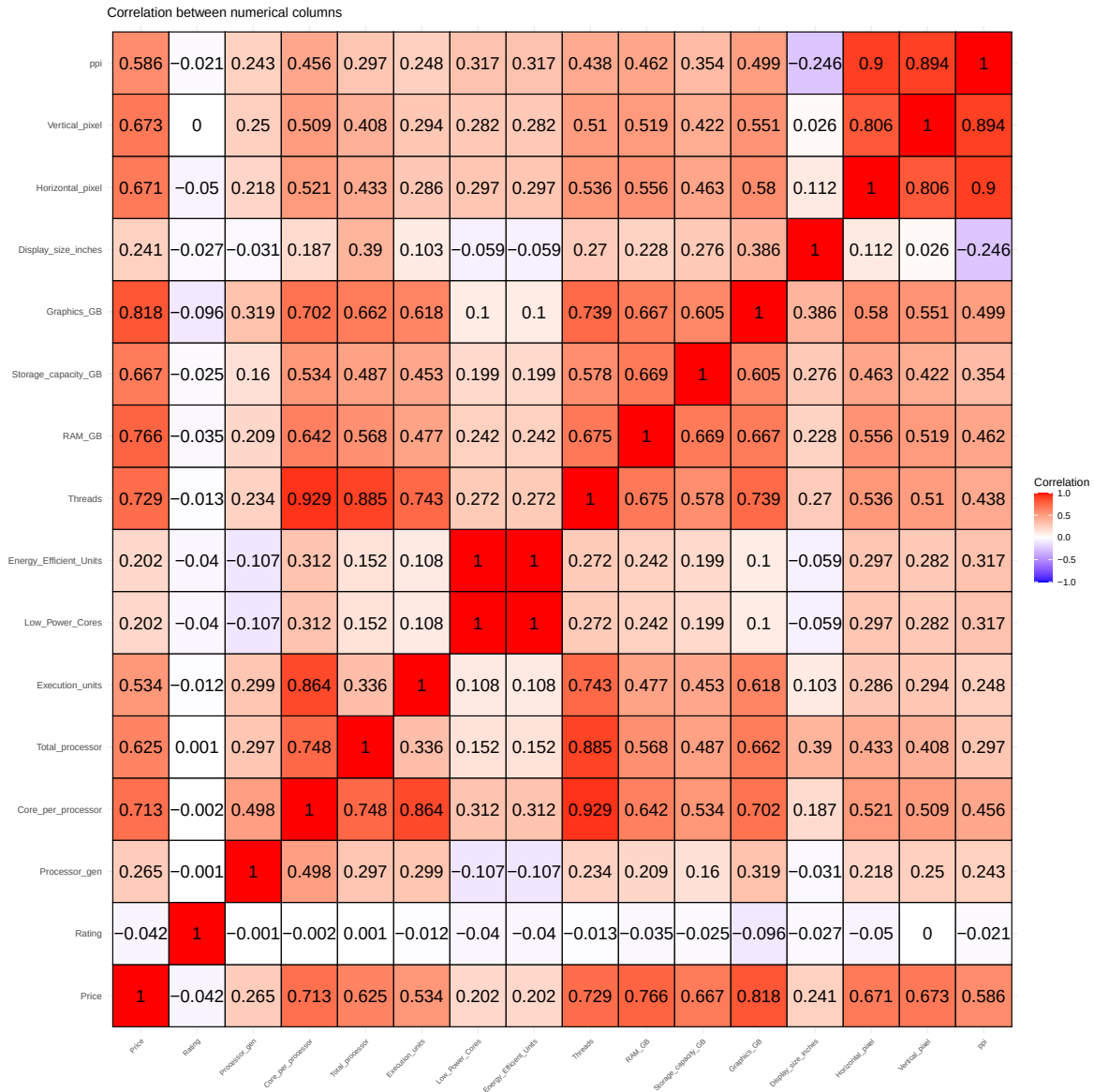
Then, look at continuous

```
library(reshape2)
library(ggplot2)
library(tidyverse)

numerical_columns <- laptop %>%
  select(where(is.numeric))
# Compute the correlation matrix using pairwise deletion.
cor_matrix <- round(cor(numerical_columns, use = "pairwise.complete.obs"), 3)

# Melt the correlation matrix for ggplot2
melted_cor_matrix <- melt(cor_matrix)

# Plotting with ggplot2
ggplot(data = melted_cor_matrix, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "black", linewidth = 0.5) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
    midpoint = 0, limit = c(-1, 1), space = "Lab",
    name="Correlation") +
  geom_text(aes(label = value), color = "black", size = 6) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1,
    size = 7, hjust = 1)) +
  labs(title = "Correlation between numerical columns", x = "", y = "")
```



```
# Delete "Graphics_GB" in the laptop data.
laptop <- subset(laptop, select = -c(Graphics_GB))
```